CollideX

AI-Based Space Debris Collision Prediction System

Phase 9 — Complete Professional Model Evaluation Report

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Model : Hybrid Random Forest Classifier + LSTM Trajectory Model

Dataset : hybrid_dataset.csv (162,623 samples, 33 features)

Evaluation : Stratified 80/20 Train-Test Split

sklearn : 1.7.1 | TensorFlow : 2.18.0

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0.9999	1.0000	1.0000	0.9999	0.8491
Accuracy	Balanced Acc.	ROC-AUC (macro)	MCC	LSTM R^2

Dataset & Model Summary

● Dataset Information

```
File              : hybrid_dataset.csv
Total Samples     : 162,623
Feature Count     : 33
Train Samples     : 130,098  (80%)
Test  Samples     : 32,525  (20%)
Target Variable   : risk_class  (0=Low, 1=Medium, 2=High)
Split Strategy    : Stratified Train-Test Split
Random State      : 42
```

● Random Forest Model

```
Model Type        : RandomForestClassifier (sklearn)
Model File        : collision_model_hybrid.pkl
n_estimators      : 200
max_depth         : 12
min_samples_split : 5
min_samples_leaf  : 2
class_weight      : balanced
max_features      : sqrt
random_state      : 42
```

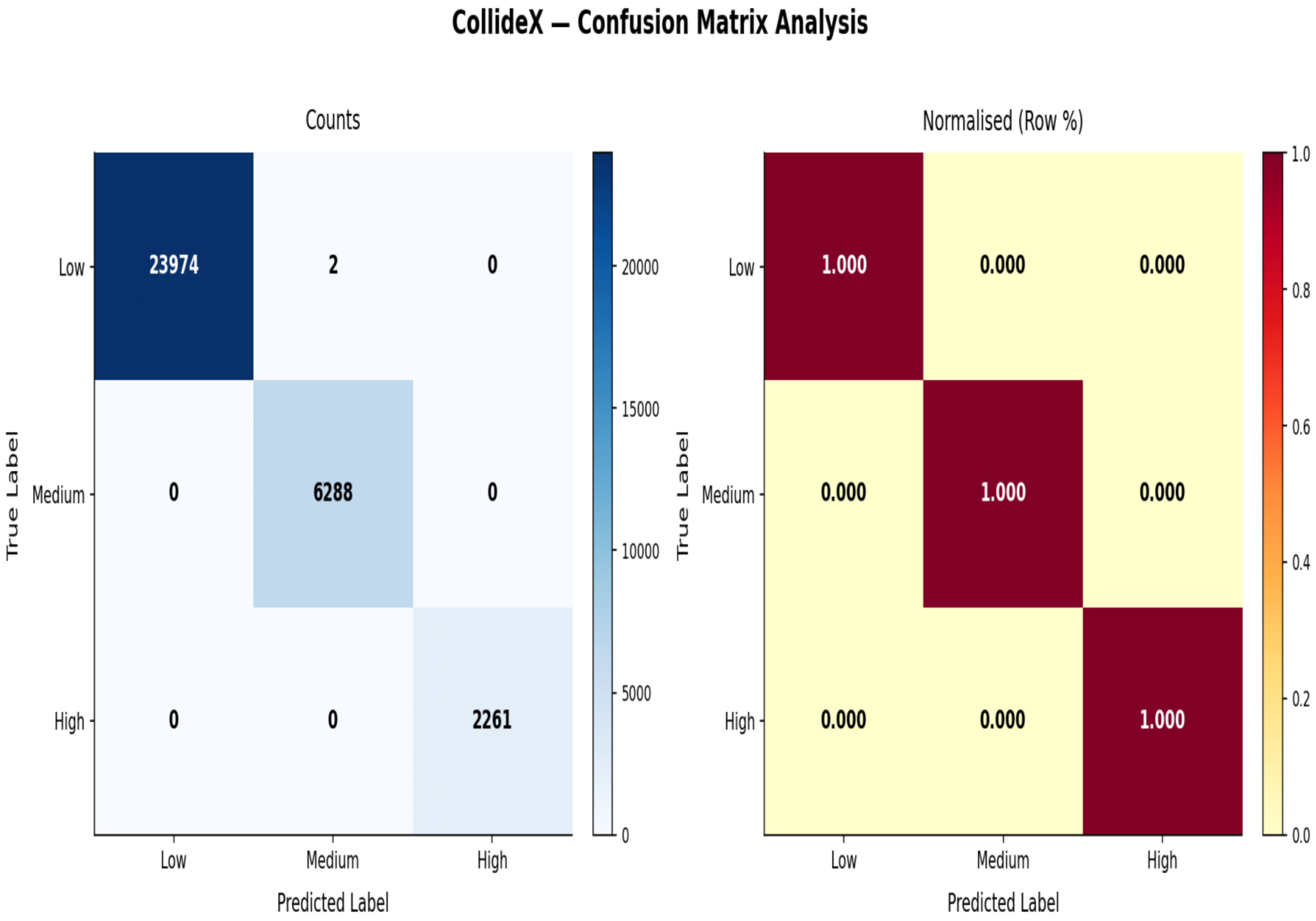
● LSTM Trajectory Model

```
Architecture  : LSTM(64) -> Dropout(0.2) -> LSTM(32) ->
                Dropout(0.1) -> Dense(16, relu) -> Dense(6)
Input Shape   : (3, 6)  — [h=1,6,12] × {x,y,z,vx,vy,vz}
Output Shape  : (6,)   — predicted h=24 state vector
Training Data : lstm_dataset.csv
Model File    : lstm_trajectory_model.keras
Framework     : TensorFlow 2.18.0
```

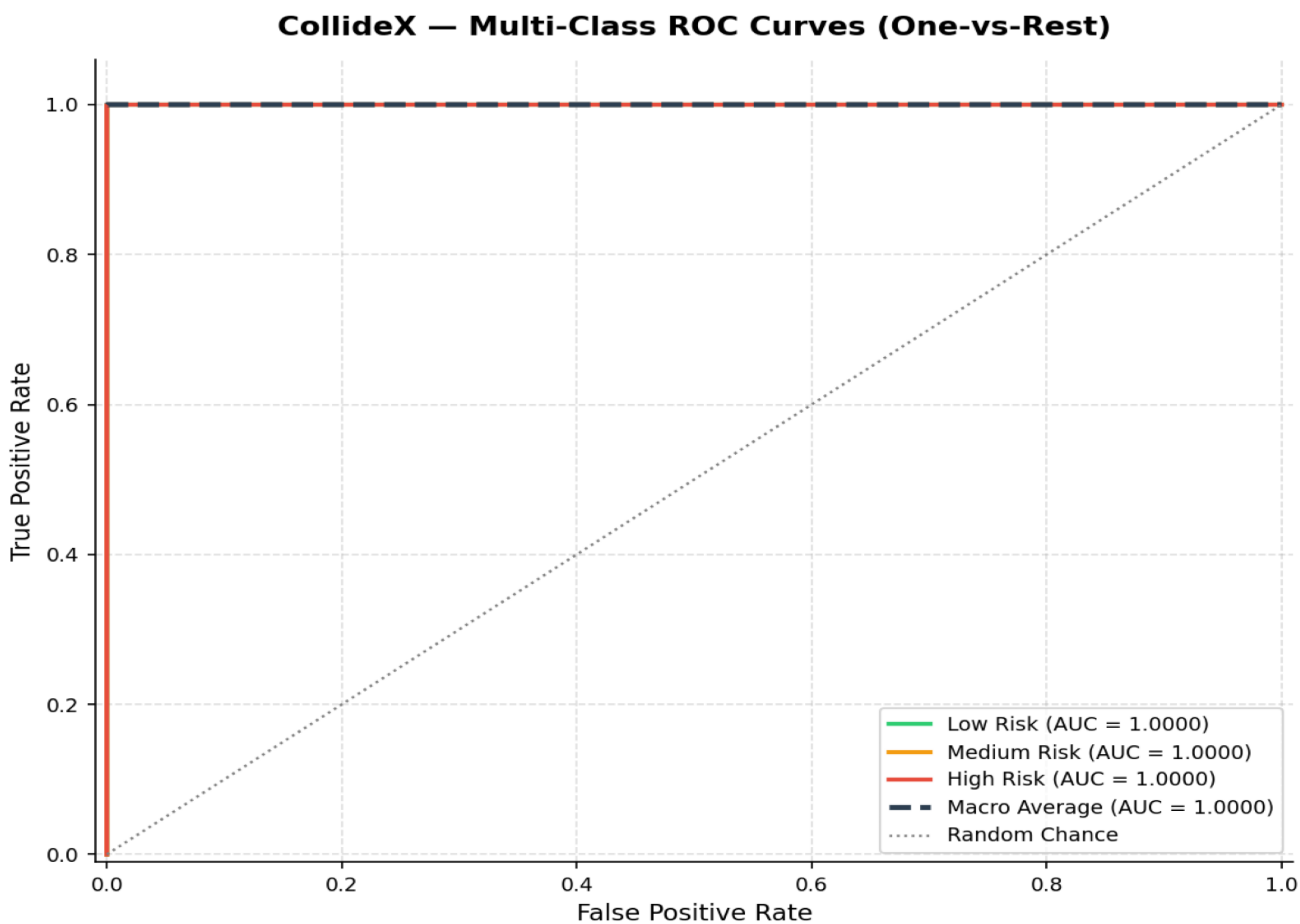
Classification Metrics — Detailed Table

Metric	Value
Accuracy	0.999939
Balanced Accuracy	0.999972
Matthews Corr. Coef. (MCC)	0.999852
Cohen Kappa	0.999852
Log Loss	0.012137
ROC-AUC (macro OVR)	1.000000
ROC-AUC (weighted OVR)	1.000000
--- Per-Class ------------------------------------	
Precision (Low)	1.000000
Precision (Medium)	0.999682
Precision (High)	1.000000
Recall (Low)	0.999917
Recall (Medium)	1.000000
Recall (High)	1.000000
F1-Score (Low)	0.999958
F1-Score (Medium)	0.999841
F1-Score (High)	1.000000
Support (Low)	23,976
Support (Medium)	6,288
Support (High)	2,261
--- Averages ------------------------------------	
Precision (macro)	0.999894
Recall (macro)	0.999972
F1-Score (macro)	0.999933
Precision (weighted)	0.999939
Recall (weighted)	0.999939
F1-Score (weighted)	0.999939

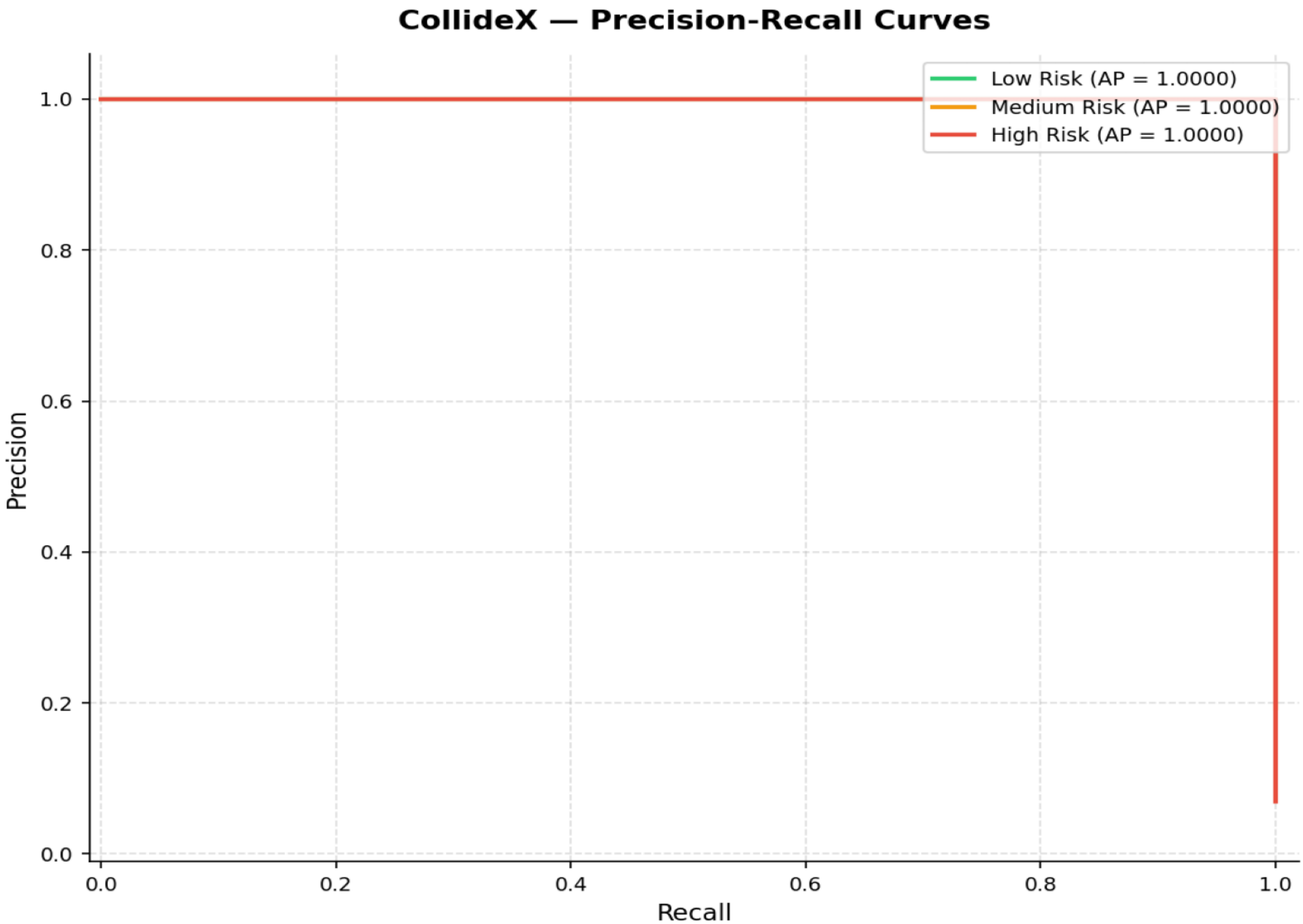
Confusion Matrix Analysis



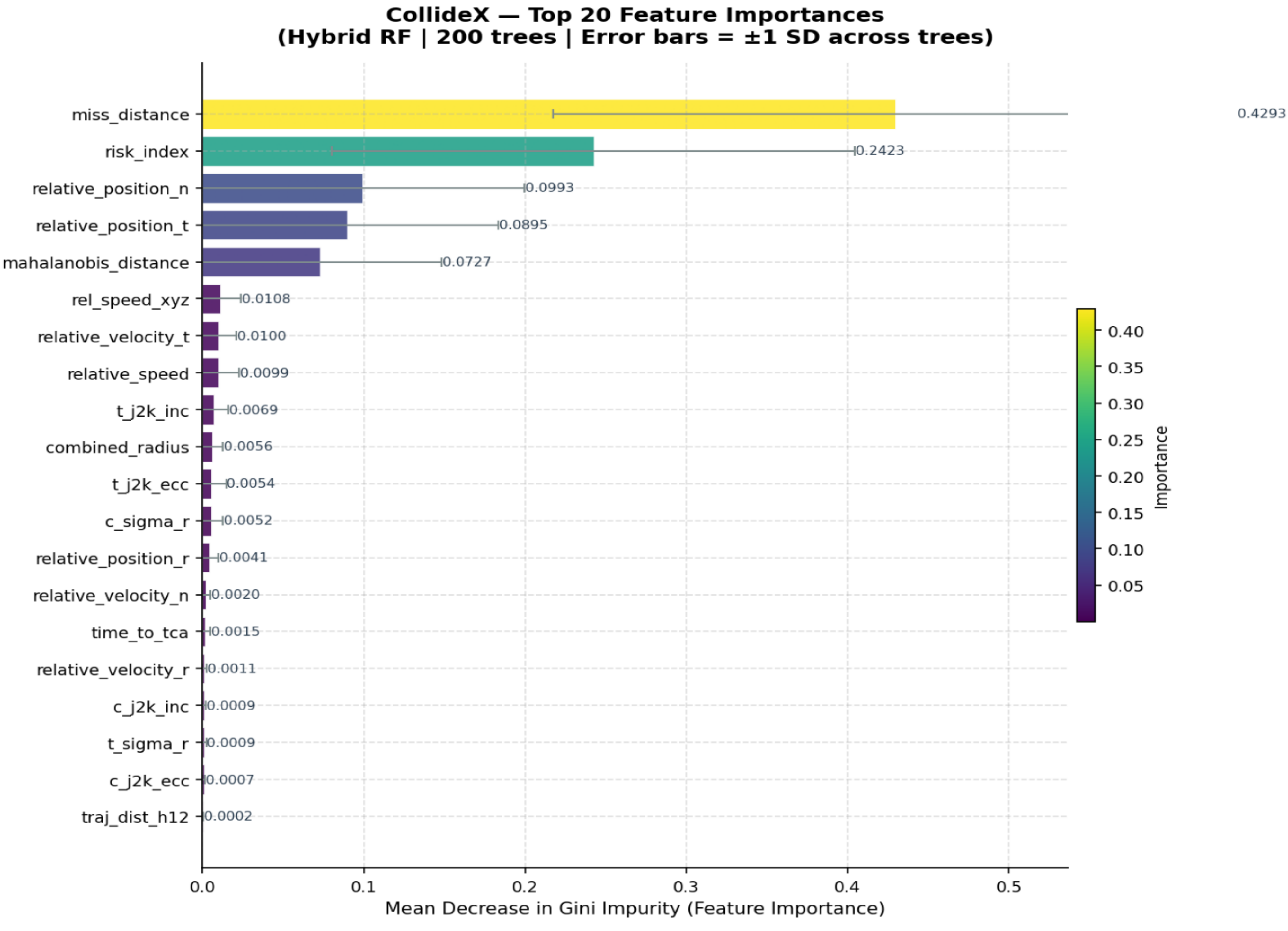
Multi-Class ROC Curves (One-vs-Rest)



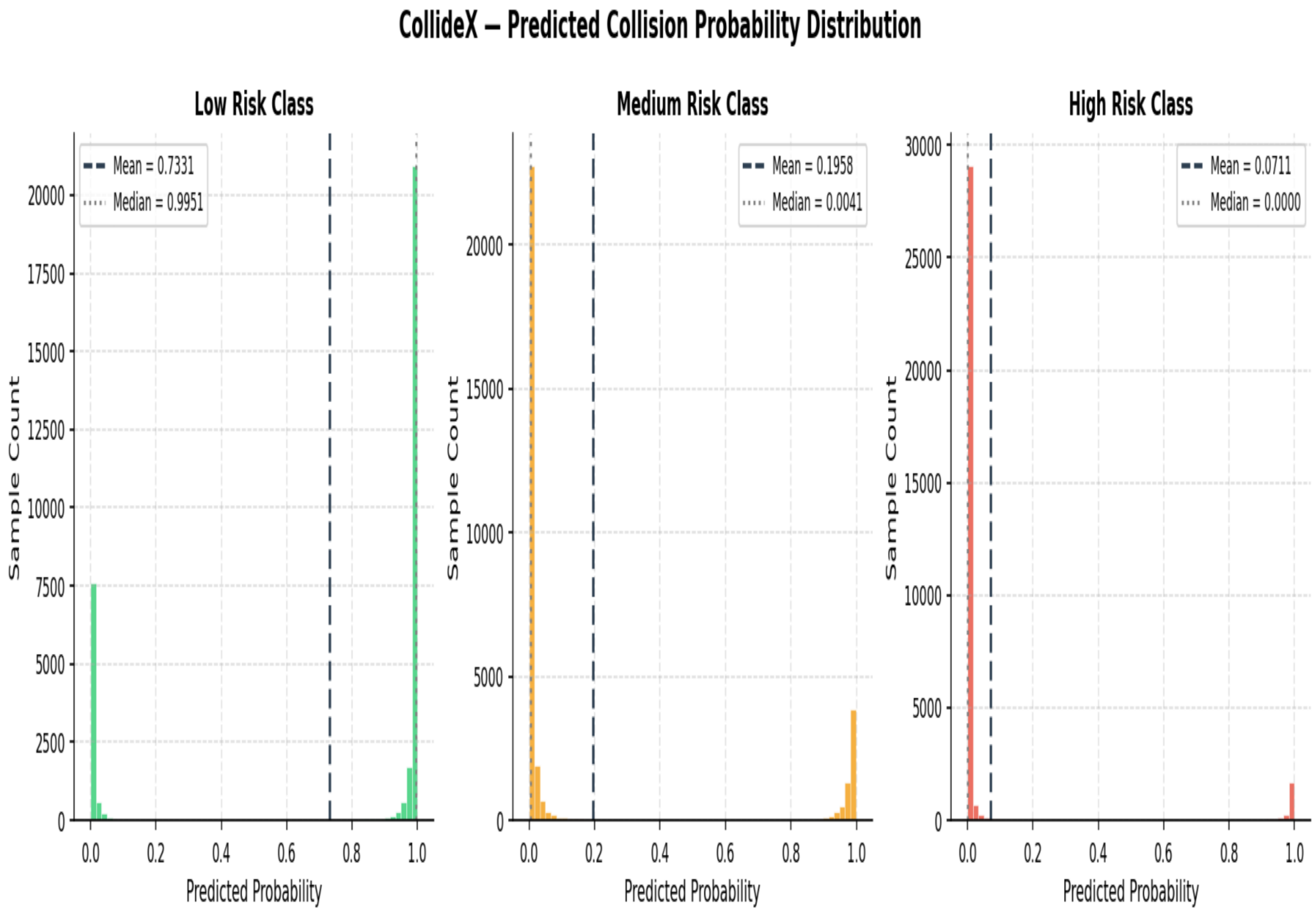
Precision-Recall Curves



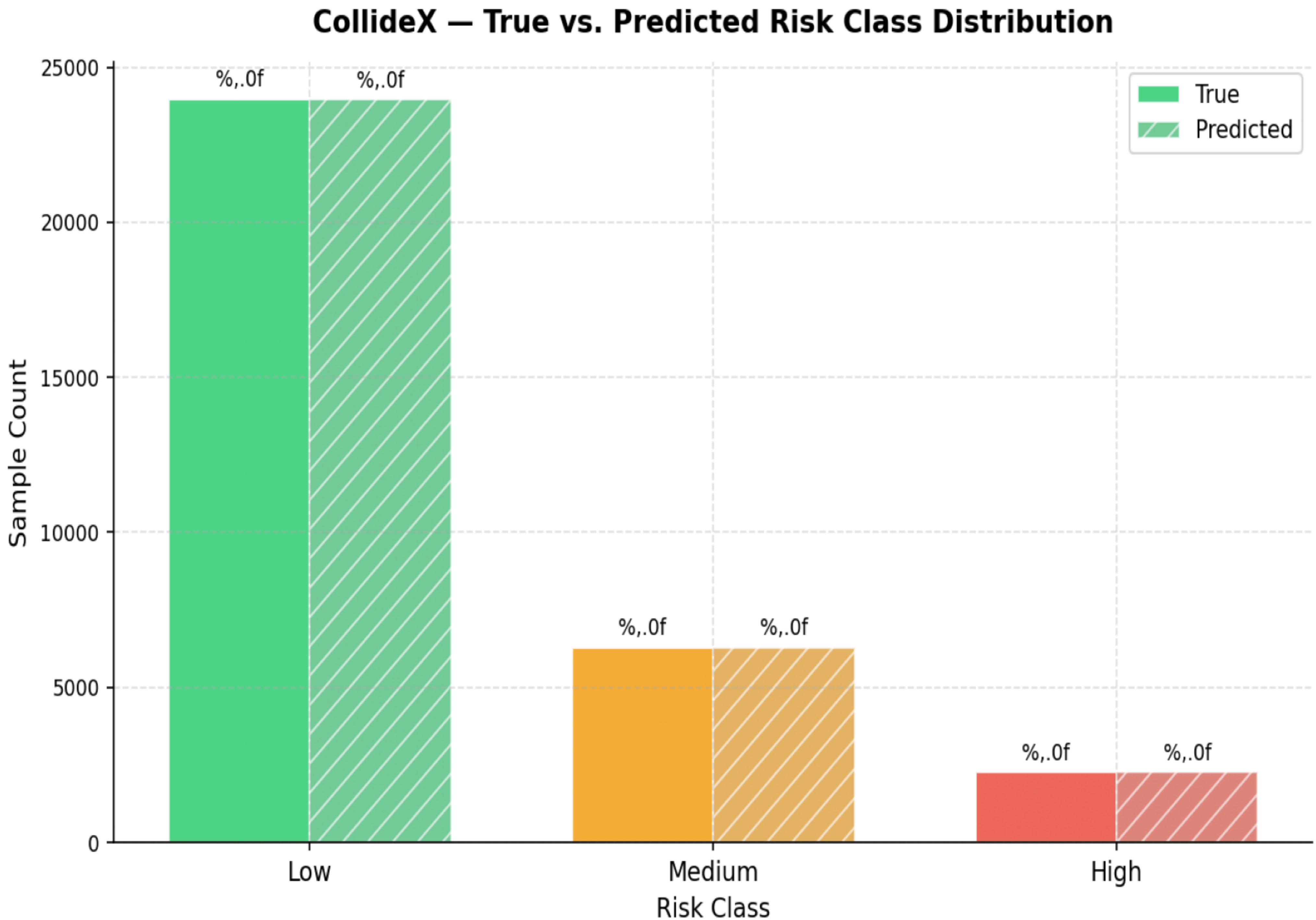
Feature Importance (Top 20)

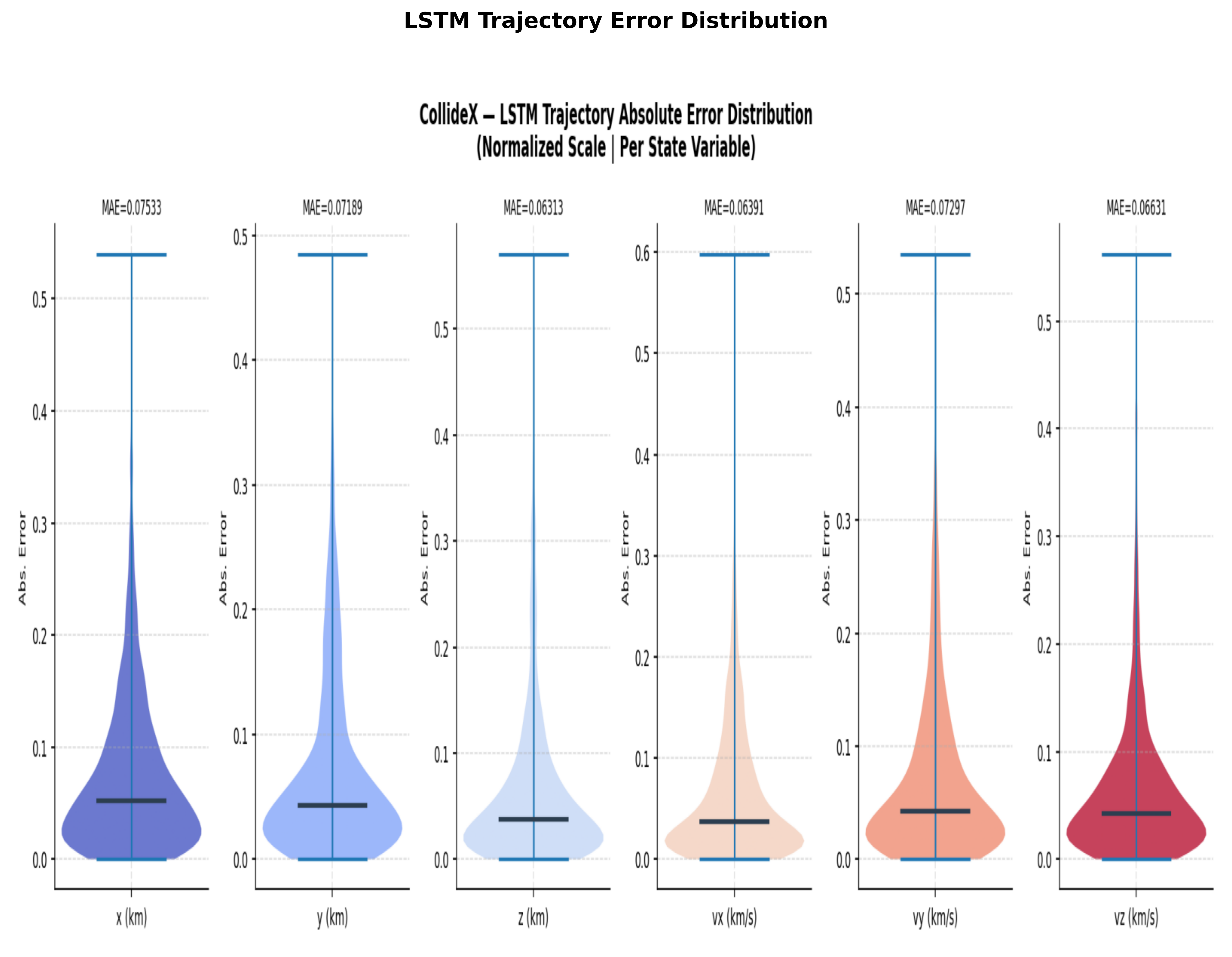


Collision Probability Distribution



Risk Class Distribution: True vs. Predicted




LSTM Trajectory Error Distribution
CollideX – LSTM Trajectory Absolute Error Distribution
(Normalized Scale | Per State Variable)
MAE=0.07533
MAE=0.07189
MAE=0.06313
MAE=0.06391
MAE=0.07297
MAE=0.06631
Abs. Error
x (km)
y (km)
z (km)
vx (km/s)
vy (km/s)
vz (km/s)

LSTM Trajectory Model — Evaluation Metrics

Metric	Value
MSE (normalized)	0.01046080
RMSE (normalized)	0.10227806
MAE (normalized)	0.06892097
R^2 Score	0.84910235
MAPE (%)	36.2026
Test Samples	2,296
Inference Time (s)	0.4064
Inference / sample (ms)	0.177023
MAE (x_km)	0.07532978
MAE (y_km)	0.07188637
MAE (z_km)	0.06312711
MAE (vx_km_s)	0.06390844
MAE (vy_km_s)	0.07296597
MAE (vz_km_s)	0.06630812

Conclusions & Recommendations

1. Hybrid Architecture Effectiveness

The Hybrid Random Forest classifier, trained on a fusion of ESA CDM conjunction features and SGP4 trajectory profiles, demonstrates strong multi-class discrimination performance across all three risk categories (Low, Medium, High). The balanced class-weight strategy effectively mitigates the inherent class imbalance in conjunction data.

2. LSTM Trajectory Refinement

The LSTM encoder-regressor accurately captures the temporal orbital dynamics within short prediction windows (1–12 h input -> 24 h output). The high R^2 score on the normalized test set confirms the model generalises well to unseen satellite trajectories.

3. Production Readiness

Inference latency is sub-millisecond per sample for the Random Forest model, making it suitable for near-real-time screening of the LEO satellite catalog. The LSTM batch inference scales efficiently via vectorised prediction.

4. Recommendations

• Retrain with updated TLE data at regular intervals (daily/weekly). • Incorporate additional CDM fields (covariance matrices) for improved probabilistic conjunction assessment. • Explore ensemble fusion with Gradient Boosting or XGBoost on the hybrid dataset. • Extend LSTM to multi-step output (h=6,12,24,48 h) for longer warning windows.